

Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Ohmic and Non-Ohmic Conductors

Three components, three shapes. Read the shape and you know the component.

Key concept

AN OHMIC CONDUCTOR IS ONE WHOSE RESISTANCE DOES NOT CHANGE

For an ohmic conductor **at a constant temperature**, the current through it is **directly proportional** to the potential difference across it:

$$I \propto V \quad \text{so} \quad R = \frac{V}{I} \text{ stays constant}$$

On an I - V graph that is a **straight line through the origin**. Double the pd, double the current. A fixed resistor at room temperature is the example.

The four words that matter: at a constant temperature. Every ohmic conductor stops being ohmic if you let it get hot.

Non-ohmic: the resistance changes as you use it

The resistance of a **lamp, diode, thermistor** or **LDR** is **not constant** — it changes with the current through the component. Their I - V graphs are **not** straight lines.

The filament lamp — and why it bends. Follow the chain, it is worth three marks on its own:

larger current → filament gets **hotter** → the metal ions **vibrate more** → electrons **collide** with them more often → **resistance increases**

So the current rises *less and less* for each extra volt, and the line bends towards the V axis — the same way whichever direction you drive it.

The diode. Current flows in **one direction only** — forward, once the pd is above roughly 0.6 V. In **reverse** it has a **very high resistance**, so essentially no current flows.

THE TRAP — and it is the one that separates grades here.

Resistance is **always** $R = \frac{V}{I}$ — and on a *curve* you must read **one point** off the graph and divide. Read V across, read I up, divide.

The gradient of an I - V curve is NOT the resistance. On a straight line through the origin the gradient happens to equal $1/R$, which is why people get away with it — but the moment the line bends, the two are different numbers. **Always go back to V/I .**

Worked example

A filament lamp is tested. At $V = 2.0$ V the current is 0.40 A. At $V = 6.0$ V the current is 0.60 A. Show that its resistance increases.

At the first point: $R = \frac{V}{I} = \frac{2.0}{0.40} = 5.0 \, \Omega$

At the second point: $R = \frac{V}{I} = \frac{6.0}{0.60} = 10 \, \Omega$

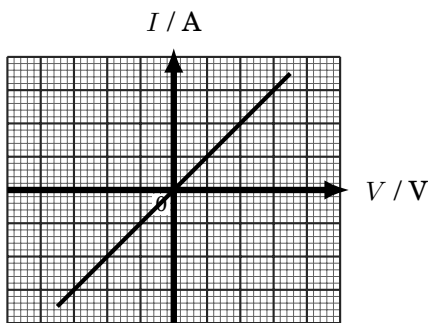
The resistance has **doubled**, because the filament is much hotter at 6 V. Both calculations used a *single point* — neither used a gradient.

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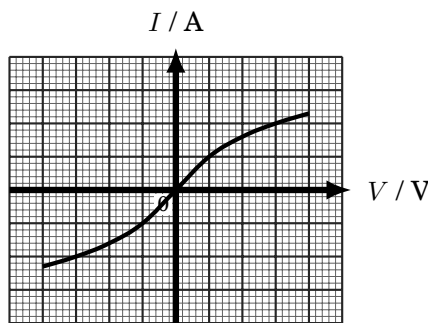
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The three characteristics

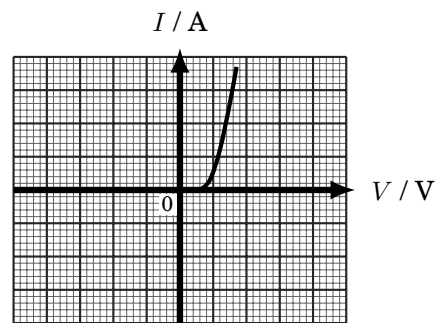
On every graph below: one large square across = 1 V, one large square up = 0.20 A.



Graph A



Graph B



Graph C

1. State what is meant by an **ohmic conductor**.

[2]

2. Name the component that produced each graph. Choose from: **filament lamp**, **diode**, **fixed resistor at constant temperature**.

[3]

A:

B:

C:

3. Which *one* of the three graphs shows a component with **constant resistance**? Explain how the shape tells you that.

[2]

4. Explain, in terms of the filament, **why** graph B bends the way it does.

[3]

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5. Use **graph B** to find the resistance of the lamp at $V = 1.0\text{ V}$ and at $V = 3.0\text{ V}$. Show your working, and state what has happened to the resistance. [4]

6. Jamie looks at graph B and says: “*The resistance of the lamp is the gradient of the graph.*” Explain why Jamie is wrong, and state what he should do instead. [3]

7. Look at the **left-hand half** of each graph, where the pd has been reversed. [4]

(a) Describe what graph C does when the pd is reversed, and explain it.

(b) Graph B is symmetrical about the origin. What does that tell you about a filament lamp?

8. A fixed resistor is left switched on until it becomes noticeably warm, and the test is repeated. Circle what happens to its I – V graph, and explain in one sentence. [2]

it begins to bend over it stays perfectly straight it flattens along the axis

9. The resistance of a **thermistor** and of an **LDR** also changes. Complete both sentences. [2]

The resistance of a thermistor _____ as the temperature _____.

The resistance of an LDR _____ as the light intensity _____.

10. Sketch the I – V characteristic you would expect for a **filament lamp**. Use both halves of the axes. [3]

